Monitoring of Inland Water Levels by Satellite Altimetry and Deep Learning

Fernando Niño, Clément Coggiola, Denis Blumstein, Léa Lasson, and Stéphane Calmant

***Abstract*— Deep convolutional neural networks (NNs) have proven their efficiency for image processing and are routinely used for image classification. In this article, we use them to convert radar measurements into water distance and ultimately into water levels of inland waterbodies. The measurements used are the successive echoes of the spaceborne radar altimeter signal on a waterbody, the radargram. We show that by using forward modeling with an accurate altimetry simulator, we can generate a sufficient amount of radargrams and train a deep NN accurately enough to obtain water level series from radargrams in a hydrology context. The method is validated at selected waterbodies by comparing these water level time series with *in situ* measurements on rivers whose width varied between 50 m and 4 km. The correlation of these time series with *in situ* data was over 0.95 with root mean square error (RMSE) between 26 and 43 cm. The results were also more robust than the Offset Center of Gravity (OCOG)/Ice-1 retracker time series of the same data. The validation shows that this automatic method performs generally as well as a carefully tuned manual method for removing outliers from the ranges provided by the state of the art classical retrackers used by the spatial hydrology community. This new tool is a big step toward a generic, global, and automated method to retrieve inland water levels from altimetry measurements. This goal is especially important in the context of continuously declining number of *in situ* measurements and of utmost importance for adequate water resources management at the global scale.**

***Index Terms*— Artificial intelligence (AI), deep learning, hydrology, Jason-3, neural network (NN), radar, satellite altimetry.**

I. Introduction

INLAND waters are an essential part of the water, carbon and energy global cycles. River and lakes water levels are essential climate variables (ECVs) as defined by the Global Climate Observing System (GCOS). From a societal point of view, surface waters are the most important source of water for domestic, agricultural, and industrial usages. Furthermore, water resources over the continents have been and will continue to be affected by climate change and anthropogenic impact [1], [2]. The past three decades were exceptional in flooding in Europe, when compared to the last 500 years [3]. In many countries, not only floods but also droughts are having a strong impact on human societies. Measuring water levels on a global scale is of utmost importance for adequate water management.

Traditionally, water level observations are performed by *in situ* gauge stations whose continuous operation are costly and not always possible in countries of political and economical unrest. The number of publicly available water level temporal series has dwindled in the last decades, making databases of water levels sparse and very heterogeneous in space and time [4], [5]. By virtue of their global coverage and continuous operation, spaceborne radar altimeters can mitigate the lack of *in situ* data [6]–[8]. Indeed, these altimeters are designed to measure the distance (or *range*) to the surface beneath the satellite. Because the satellite position is known very precisely, when it flies above the ocean, a river or a lake, one can obtain the water height by subtracting the altimeter range from the satellite altitude (and other terms to account for propagation delays and Earth deformation, which will not be discussed here). To make possible the monitoring of geophysical quantities, altimetry satellites are placed on repeat-track orbits; this means that the radar altimeter will pass at regular intervals over a waterbody and can provide time series of water levels with a sampling period equal to the cycle duration of the orbit (ca. ten days for the Jason-3 satellite).

In the following text, radar altimetry data will be used to estimate the water level of inland water bodies, and obtaining water level time series. We will call *hydrology targets* the intersections between satellite groundtracks and waterbody centerlines (Fig. 1). By analogy with the *in situ* stations, hydrology targets will be called *virtual stations* when it is possible to extract good quality temporal series of water levels from the altimetry measurements. For validating our work, we will use mainly water level data from *in situ* stations, but we will also compare our results to the standard estimation method of the Offset Center of Gravity (OCOG) retracker (cf. Section III).

As formulated above, obtaining temporal series of water levels from space seems straightforward; however, for over 20 years different research teams have been working on this problem, and the number of available virtual stations is only a small fraction (less than 10%) of the available number of hydrology targets. There are at least two explanations for this low ratio [9].

1) Radar altimeters, which were designed primarily to measure the oceans, can have difficulties to keep the radar echo on the acquisition window (or to have correct *tracking*) when faced with the rapid change of radiometry and topography over lands [10].

Manuscript received June 16, 2021; revised September 16, 2021 and November 29, 2021; accepted December 21, 2021. Date of publication December 23, 2021; date of current version March 3, 2022. This work was supported by the Centre National d'Études Spatiales (CNES) through the VOLODIA Project and the OSTST/Tosca Program. *(Corresponding author: Fernando Niño.)*

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Digital Object Identifier 10.1109/TGRS.2021.3138329

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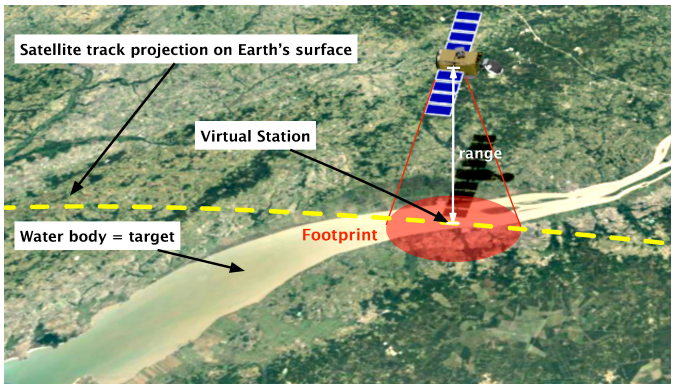


Fig. 1. Figure showing the terminology used: the hydrology target is the water body, the virtual station being the intersection of the waterbody and the satellite track projection. The distance between the satellite altimeter and the target (waterbody) is the range. The radar altimeter echo does not bounce off a single point, but integrates the echos of a region, called the footprint.

TABLE I
VIRTUAL STATIONS AVAILABLE IN ALTIMETRIC HYDROLOGY DATABASES

Database Name	Virtual Stations	Target %
HydroWeb	12966	7.2
DAHITI	2894	1.6
Hydrosat	1714	1
Total	**17816**	**9.8**

TABLE II
HYDROLOGY TARGETS FOR ALTIMETRY MISSIONS USING THE DEM TRACKING MODE

Mission	Hydrology Targets
Jason 3	33445
Sentinel 3A	74051
Sentinel 3B	73630
Total	**181126**

2) The validation of altimetry measurements over inland waters requires a great deal of expertise to remove the frequent outliers.

The first problem has been tackled in the last years by the operational use of the new onboard digital elevation model (DEM) tracking mode (also called *Open Loop mode*) implemented on recent altimeters (Poseidon-3 on Jason-3, Sentinel-3 Radar ALtimeter (SRAL) on Sentinel-3) [11]–[13]; for past measurements, wrong onboard retracking means the radar echoes are unfortunately lost, and for these cases, the retrieval of altimeter range is impossible. Hopefully, other echoes corresponding to the "wrong" object will be detected as such.

The methods proposed in this article aim at solving the second problem by providing automated methods to convert altimetry measurements into range over inland waters. This automation is even more important when confronting the task of processing the increasing number of altimetry measurements in virtual stations made available by the onboard DEM tracking mode (compare the availability of virtual stations of Table I with the number of hydrological targets attainable for missions using the DEM tracking mode in Table II). In this article we will focus on the Jason-3 mission because of the availability of this DEM tracking mode, but the methodology can be applied to other missions.

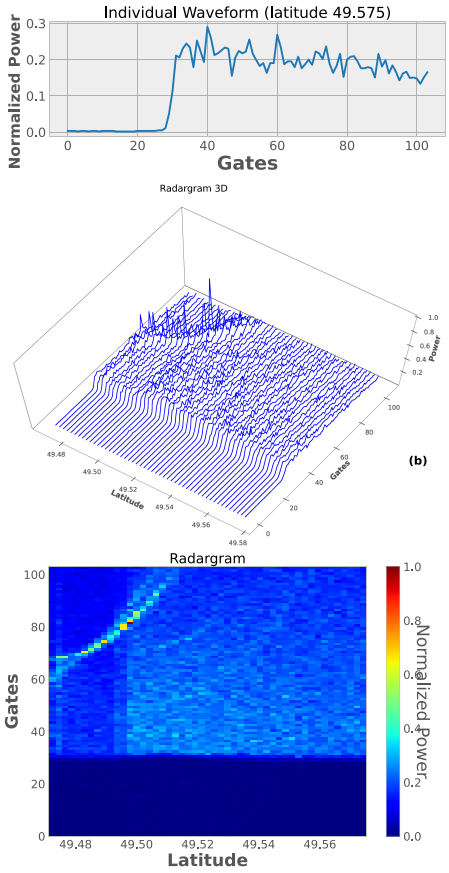


Fig. 2. Radar altimetry echoes using data of pass number 146 of Jason-3 on the North Atlantic on April 2, 2016 (see text for more information). (a) View of one waveform, showing the evolution of measured signal power through time, as sampled in the so-called gates. (b) 3-D view of a sequence of echoes, with axes latitude, gates, and power. (c) Radargram, which is a view of (b) from above, where power values (heights) are color coded.

Radar altimeters do not measure directly a range value at each point, but the echo of a radar signal (or *waveform*) from which the range value must be retrieved. Radar echoes are processed onboard and transmitted to the ground at a relatively high frequency (20 Hz for Jason-3), which is equivalent to a spatial sampling every few hundreds of meters along-track. If we juxtapose consecutive altimetry waveforms along the time axis (or equivalently, by latitude), this waveform sequence can be viewed as a small grayscale or pseudocolor image of reflected power. These images are called *radargrams*. In Fig. 2, we show various representations of radar altimetry echoes for data corresponding to pass number 146 of Jason-3 on the North Atlantic on April 2, 2016, just offshore Normandy in the north of France. Far from the shore, one radar echo [Fig. 2(a)] behaves as expected from the Brown model, as discussed in Section II. This is a time versus returned-power plot, where the time axis has been converted to the equivalent gate samples. Fig. 2(b) shows the 3-D view of a radargram and Fig. 2(c) a pseudo-color plot of the radargram. For this particular case, the sand on the nearby beach appear as a bright parabola segment on the radargram, just as a river would stand out on a low reflectance landscape.

With some expertise, it is easy to recognize in the radargrams the signature of different objects (e.g., the parabola in Fig. 2). This expertise being essentially visual, we have

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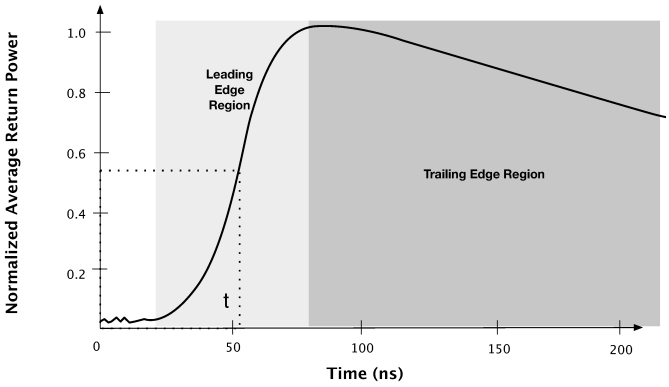


Fig. 3. Simplified schema of the Brown model for ocean waveforms. The midpoint of the leading edge of the waveform (epoch, or τ) corresponds to the range of the altimeter for this echo.

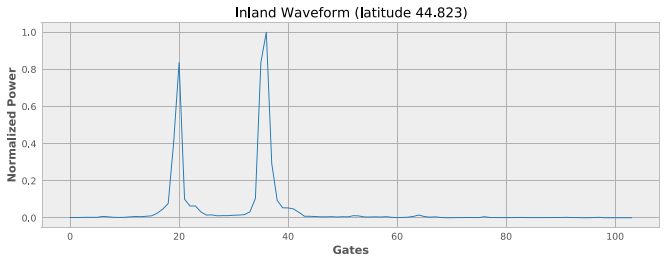


Fig. 4. Example of inland waveform not correctly described by the Brown model (cycle 18, pass 35 of Jason-3 overpass of the Dordogne river in South Western France, latitude 44.823, longitude 0.134). Two peaks show the ambiguity of deciding which peak corresponds to what waterbody below.

used supervised learning to train a deep convolutional neural network (NN) [14], [15] to learn the function that identifies the correct value of radar range (and hence, the water level) that is to be associated with a waterbody, by analyzing its radargram.

The rest of the article is structured as follows: in Section II, a review of radar altimetry waveform properties is presented. We also highlight the differences between ocean and inland water processing for radar range retrieval, and how a radargram is used to compute the water height of a waterbody. Section III presents a summary of the alternative strategies put forward for retrieving the altimeter range of waterbodies, Section IV describes the methodology used to construct the NN. Section V shows the experimental results and their validation with respect to *in situ* data. We close this article with a discussion on NNs for hydrology processing, possible follow-ups, and conclusions.

II. Waveforms in Satellite Altimetry

Satellite altimetry was designed for measuring the sea surface with a radar signal. Brown [16] built a theoretical model of the response of the radar signal to an ocean-like surface (a surface whose heights follow Gaussian statistics). A simplified schema of the Brown model is shown in Fig. 3. By fitting the observed waveform to this theoretical model, one can estimate several parameters (altimeter range, backscatter coefficient, satellite mispointing) for oceanography applications. Of these, we will only be concerned with the altimeter range.

For waveforms of a form such as that of Fig. 3, the altimeter range will be obtained by the expression

$$\text{range} = r_{\text{tracker}} + (\tau - \tau_{\text{ref}})\delta r \tag{1}$$

where r_{tracker} is the one-way distance between the center of mass and a fixed reference point on the sampling window, τ_{ref} (gate 32 for the Jason-3 altimeter). The epoch of the waveform, τ, is the time associated with water height (the midpoint of the leading edge in Fig. 3). The conversion of gates to distance is made by factor δr, which is the radar distance resolution (ca. 47 cm for Jason-3).

As shown in Fig. 1, the radar echo is not the result of the interaction of the radar signal with a point on the surface, but with a *footprint* region. This footprint is just the region located at such a distance that the radar signal reflected by the ground reaches the receiving antenna in the expected time window to contribute to the waveform. In the case of the ocean, this footprint is approximately a circle. Ocean water height varies with approximately random Gaussian statistics, where the Brown model applies very well. For the case of inland water this is clearly not the case; land embedding the waterbody is generally not described by random heights and thus the Brown model does not apply. Moreover, a radar waveform of the ocean type fits easily to the Brown model and presents no ambiguity to define a range: there is a clear maximum that separates a leading edge and a trailing edge. By contrast, as shown in Fig. 4, multiple peaks can appear on an inland waveform, which may correspond to different waterbodies (ponds or swamps for instance), and not necessarily the main hydrological target. However, if we look at its corresponding radargram (Fig. 5), we see that high reflectance bodies (e.g., water on a low reflectance background such as cropland) appear as parabolas on the radargrams; this is also known as the snagging issue. A simple geometrical calculation shows that a bright object on the ground is visible on several waveforms and the loci of the maximum of these waveforms in the radargram follow a parabola (or the branch of a hyperbola, strictly speaking). The epoch of the radargram will be the gate coordinate corresponding to the lowest point in the parabola associated with the main hydrological object (black line in Fig. 5).

The main insight is thus that bright waterbodies affect several successive waveforms in the neighborhood of the one corresponding to nadir, and that the estimate of their height and location can benefit of a radargram analysis, instead of relying only on individual waveforms.

Efforts have been made to automate this process, and our hydrology group has been using semiautomated software tools for over a decade: Virtual ALtimetry Station software (VALS) [17], Multi-mission Altimetry Processing Software (MAPS) [18], and more recently ALtimetric TIme Series software (ALTIS) [19] in which the operator must manually select the measurements to take into account when building a time series, eventually taking into account the slant ranges.

When the number of available hydrological targets is in the hundreds of thousands (since this analyses must be made for every measurement over a particular hydrological target), this approach loses its appeal, and further automation is required. Radargram parabola analysis can be automated, but our tests with this approach showed that only ca. 30% of the cases can be treated algorithmically, many fuzzy cases remaining intractable.

Rather than complexifying the algorithmic approach to take into account more specific cases, we propose a machine

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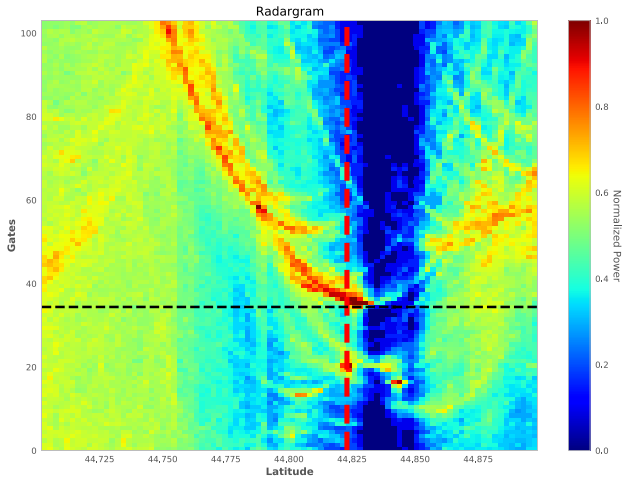


Fig. 5. Inland waters radargram (to make features stand out, we plot here the logarithm of the echo signal). The position of the waveform of Fig. 4 is shown with a dotted red line. The parabolas appear as a consequence of the high reflectance of waterbodies, which appear on different waveforms as seen at different angles and hence, distance. The horizontal black line is a representation of the epoch of the radargram, which in this case corresponds to the ordinate of the lowest point of the strongest parabola.

learning approach in which a NN is able to return the range of the main waterbody given its radargram for the general case of hydrology targets.

III. Related Work

For hydrology applications, altimetry has the appeal of global coverage, insensitivity to cloud cover, and good accuracy. Many approaches have been taken to exploit these advantages for obtaining hydrological time series from altimetry waveforms in continental surfaces.

As mentioned in Section II, altimetry was designed for ocean surfaces, and the waveforms obtained in hydrology contexts are not compatible with ocean waveform-processing algorithms. The processing algorithms that take waveforms as input and obtain the corresponding range (among other parameters) are called retrackers. For ocean surfaces, retrackers are based on fitting the Brown model, and operational processing chains use a maximum likelihood estimator with three or four parameters. The topographic configurations in which inland water bodies appear have great diversity, and the radar altimetry waveforms can thus be very different from one another, having one, or many peaks and no clear standard. Many retrackers have been developed to cope with this situation.

Koblinsky *et al.* [6] were one of the first to study great river basins, and proposed two different models for Geosat waveforms following the work of Zwally *et al.* [20]: a five-parameter single-ramp similar to the Brown model, and a nine-parameter double-ramp function. These models were to be chosen according to a simple classification: if the waveform was diffuse, the single-ramp model should be used; if it was specular, the double-ramp model should be preferred. The results they obtained were not adequate for routine hydrological measurements because of their accuracy and the problems with the Geosat altimeter, particularly the uncertainty in its orbit. Birkett [7] constructed one of the first accurate lake level datasets, showing the usefulness of Topex/Poseidon altimetry in a hydrological context. Rather than developing a new retracker, she constructed a time series of height data relative to a reference pass profile, whose only constraint was that its slope should not exceed 0.25 m/km. The dataset consisted of 24 big lakes, with most smaller lakes being missed because of the great inter-track distance of Topex/Poseidon, and the effect of topography variations, which hindered radar echo acquisition (problem now solved with on-board DEM tracking mode) and low accuracy due to the quality of the atmospheric corrections. Birkett [8] later extended her analysis to eight large rivers (width greater than 2 km) and nine wetlands. Multimission altimetry data ([9], [21]) were also successfully used for monitoring big rivers and supporting the case for global hydrology.

Frappart *et al.* [22] analyzed the four different retrackers used in the ENVISAT mission's official products: an ocean retracker (based on the Brown model, and using Hayne's expressions [23]), the OCOG (or Offset Center of Gravity, or Ice-1) retracker [24], the Ice-2 retracker [25], and Laxon's Sea-ice retracker [26]. Remarkably, none of these retrackers was created for hydrological applications. Two of them were refinements of the Brown model (ocean and Ice-2), the other two being intended for ice applications. OCOG calculates the center of gravity, amplitude, and width of a rectangular box using the maximum of waveform samples, and finds the point on the waveform where the amplitude exceeds a threshold. As such, the OCOG retracker does not care too much about the precise position and shape of peaks of the waveform and thus is very robust. The Sea-ice retracker is just finding, by interpolation, the point on the waveform where the echo is greater than a threshold corresponding to half the waveform amplitude. The comparison of these four retrackers showed that despite its simplicity, the most accurate water stages were obtained when using the OCOG retracker.

Following the pioneer work of Berry [9], the Pistach project [27], [28] proposed the use of a waveform classifier prior to retracking altimetry data in coastal and hydrology contexts. According to the class the waveform was assigned to, among the 16 they defined (1: Brown echoes, 2: Peaky echoes, 3: very noisy echoes, 4: linear echoes, etc.), a particular retracker was invoked. The results were promising but were never compelling enough to be of widespread use, the main problem with their product being the lack of along-track consistency when changing from one retracker to another.

A different approach for constructing time series is that of the Database for Hydrological Time Series of Inland Waters (DAHITI) database [21] that uses automatic outlier rejection and Kalman filtering to use different tracks and missions into consistent water-level series estimations. The dataset is promising, but the number of virtual stations is clearly lacking with respect to the available hydrology targets (cf. Table I).

The approach we present is much in the spirit of that of Enjolras and Rodriguez [29], but the technical details are completely different, and with outcomes greatly enhancing usability for hydrologists. In particular, we deal with global adjustment of virtual stations while they inverted the waveforms of a selected site.

With regard to recent studies of machine learning applied to hydrology, we can cite the M.Sc. of Mattes [30], which used a one-layer NN with 10 or 30 neurons for creating water level

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time series of inland water bodies. However, Mattes's network was applied on individual waveforms and not in radargrams and will probably suffer the same drawbacks on that other traditional methods that lack the context that is present on radargrams. Memarian Sorkhabi *et al.*'s recent work [31] tackles the much simpler approach of modeling the water level of one particular lake (Lake Urmia, in Iran/Azarbaijan). They apply several filtering techniques to test the performance of a convolutional NN for retracking wavevorms (again, not radargrams). Several scenarios for NN training were made, in which a different percentage of the training dataset underwent wavelet filtering before being used as input for the NN. The results were tested against *in situ* data and compared to gravity (GRACE) data. Results are good for the very specific case they are studying, with root mean square error (RMSE) of ca. 3 cm. We are not aware of any more uses of machine learning to tackle the problem of creating hydrological time series through radargrams. In this study, we do take advantage of the advances in feature extraction and image processing with deep learning to take advantage of the context contained in radargrams to give a more appropriate interpretation of the waveforms in a hydrological context.

IV. Methodology

The problem of determining inland waters level from altimetry data can be posed in different ways. Here we suppose that waterbodies (lake, river, etc.) of interest are known and that the radargrams of the regions in the vicinity of these waterbodies are readily available.

We want to build a NN that, given as input a radargram over a particular waterbody, returns the epoch corresponding to the waterbody's surface height [as in (1) and Fig. 3]. By feeding to this NN a set of radargrams of that waterbody, one for each pass of the altimeter over it, we can then construct a water-level time series. In practice, we will be using radargrams of 104 gates $\times$ 80 waveforms, the number of gates being a characteristic of the radar altimeter used, and 80 because the radargram was wide enough to fit all the observed parabolas. For doing so, we will create a deep NN by supervised learning. This technique requires a great number of training examples, defined by the couples (input = radargram, output = epoch). The number of examples required to train a deep network (more than 710 000 examples in our current training set) is much greater than the number of the *in situ* measurements available. Instead, we choose to use radargrams produced by an altimetric waveform simulator with which we control the "ground truth," since its input is the description of a scenery as defined by two maps: surface heights and backscatter coefficients. Given these maps, the simulator is able to compute the response of the scene to a radar altimeter, and produce the simulated radargrams. These radargrams can, in turn, be used by a retracker to compute the surface heights below (which are nothing else than the input surface heights). Section IV-A describes the model used for these simulations.

A. Simulations

1) Radar Altimeter: A radar altimeter can be schematically described as an instrument emitting an instantaneous radar signal and recording the echos returned from the target surface below during a very short time (roughly the time needed for the speed of light to travel a distance of 50 m, in the case of the Jason-3 altimeter considered in this article). This recording is the waveform described in Section I, and is discretized in gates (104 for Jason-3): the width of each gate corresponding to a distance δr of approximately 47 cm. Because the radar altimeter travels onboard a satellite, it moves at a ground speed of ca. 7 km/s. With Jason-3's sampling of 20 Hz, waveforms are spaced every 350 m.

A complete characterization of the radar instrument is beyond the scope of this article, but we require its position at every measurement point (given by precise orbit calculations for real altimetry missions), its bandwidth, carrier frequency, antenna diameter, point target response (PTR), and so on.

2) Scene: To simulate the formation of an altimetry waveform we consider a *scene* in which all the scattering objects are placed. This scene includes a description of the topography, rivers, lakes, urban areas, grass, sand, snow, and so on. For our simulation, this scene will be discretized with a grid of 20 m resolution.

The portion of the scene that interacts with the radar signal can be seen as a set of elementary scattering reflectors. This scene itself can be described by the positions of these reflectors and their backscattering characteristics.

If we consider a single-point reflector P receiving a radar signal from a radar altimeter at position R, then let us denote the Euclidean distance between them as $d(P, R) = r$. If we neglect the radar signal delays due to atmospheric effects, the echo created by P appears in the waveform as a peak at position g given by

$$g = (r - r_0)/\delta r \tag{2}$$

where r_0 is the distance from P to the first waveform gate and δr the gate width.

This energy peak contributes to different gates for successive waveforms. When P is at the nadir of the antenna, g is minimum and the peak is maximum. As P moves away from the nadir, g increases (late arrival) creating the parabolas in the gate/latitude plane shown in Figs. 2 and 5. The contribution of the peak decreases laterally according to distance and the antenna pattern.

A very important point here is that peak position depends on the distance between the altimeter and the reflector. A consequence of this observability issue is that if we suppose that the reflector is directly underneath the satellite, we can deduce the height of the reflector; however, if it is at a distance from the ground track, the deduced height will be wrong unless there is *a priori* information on the horizontal position of the object (for example, a water mask or optical image of the zone).

3) Backscatter and Antenna Pattern: The backscatter coefficient σ_0 is a measure of the reflecting power of the surface and is expressed in decibels (dB). The higher the coefficient, the stronger the echo. On the other hand, this coefficient depends on the angle of the surface with respect to the incident radar signal θ. Because our applications concern hydrology, the reflectors we work with are water and the normal direction to them are in general vertical.

We consider two scattering behaviors (Fig. 6).

a) *Diffuse or Lambertian Reflector:* No angular dependence, σ_0, is independent of θ.

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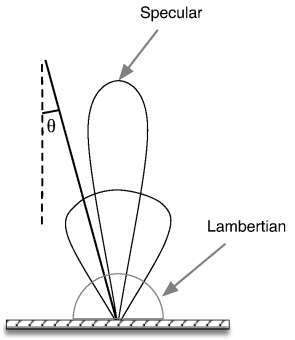


Fig. 6. Diagram showing three scattering behaviors for a scene element: diffuse (Lambertian), specular, and a mix of both.

b) *Specular Reflector (Mirror):* Very marked dependence on the incidence angle θ. The function $\sigma_0(\theta)$ has a strong peak near the vertical, for value $\theta = 0$. The specularity function is modeled as $e^{-g_0\theta^2}$. This Gaussian function is parametrized in terms of its full width at half maximum, fwhm.

In reality, a mix of both behaviors is frequently observed. The fraction of diffuse reflectance in the mix is called x_{diff}, and that of specularity $(1 - x_{\text{diff}})$.

The radar echo is captured by an antenna whose performance is crucial for measurement acquisition. The antenna pattern is defined by the gain $G(\theta)$ with which the radar power is damped at reception. $G(\theta)$ is modeled by a Gaussian function with a fwhm value of 1.3°, typical of most altimeters.

4) Simulator: Putting all this together, the power bounced back by the reflector P from the signal of the altimeter at position R, and received at gate g (as defined above) is

$$dp(g(P, R)) = A \cdot G(\theta(P)) \cdot \sigma_0(P, \theta(P)) \cdot dS \tag{3}$$

where dS is a surface element associated with P and A is a scaling coefficient, which takes into account many system parameters, such as the emitted power, the electronic gain of the instrument, and the free space losses.

The simulator then models the construction of a complete waveform by integrating the contributions of all the grid surface using the following equation:

$$p(g(P, R)) = \int_S A \cdot G(\theta(P)) \cdot \sigma_0(P, \theta(P)) \cdot dS. \tag{4}$$

It is to be noted that this simulation makes use of the incoherent sum of the individual reflector echoes, as it was done in [29]. A more complete approach is also possible [32] in which the waveform is constructed by a coherent sum of the contributions of the reflectors (i.e., taking into account not only the amplitude of the received signal, but also its phase).

The variability of the simulated radargrams depends both on geophysical variations (the landscape changes through time, as well as weather conditions) and on differences in the location of altimetry measurements. Even if the satellite altimeter follows a repeat orbit, there are variations on the actual trajectory, which can wander up to 1 km (cross-track distance) away from the theoretical ground path (orbital maneuvers conducted by the space agencies correct these drifts). For hydrological observations, the radargrams will not always have the targets at exactly the same position. The altimeter acquisition window (when the recording of the echo of the emitted signal is made) is defined on-board and is used to define the distance of the altimeter to a reference gate in the radargram (the r_0 of (2), whose value is 32 for Jason-3). This distance is the tracker range.

It is worth to note that other technical details must be accounted for in order to obtain realistic waveforms: such as pulse compression for the emitted signal (since the signal is not infinitely short), variable bandwidth for some altimeters (e.g., the ENVISAT mission's RA-2 altimeter changes resolution depending on the observed surface). The simulator takes them also into account, but a technical presentation of these features are not relevant for the analysis that follows.

The simulation of realistic waveforms requires not only good knowledge of the technical details of the altimeter instrument but also good ancillary data of the scene and of the instrument (tracker range and its automatic gain control, or AGC). Surface heights are derived from Shuttle Radar Topography Mission (SRTM)-30 [33], a global DEM; backscatter characteristics for hydrology can be defined by water masks (e.g., [34], [35]) using the fact that the reflectivity of inland water is significantly higher than that of the land surrounding it, for the case of the Ku-band used here. We hereto describe how the water reflectance is constructed when preparing the simulations of the machine learning database.

The radargrams in the database are generated in several steps, and satisfy several conditions.

1) Locate the virtual stations (intersections of Jason-3 ground tracks with known waterbodies from the SRTM Water Body Data (SWBD) water mask database [35]). A map of the 2517 selected virtual stations is shown on Fig. 7. Each virtual station will be used at least as many times as there are flyovers using DEM tracking (later than September 2017, on cycle 58).
2) A scene is created using a realistic DEM around each virtual station using the SRTM-30 dataset [33]. The scene is typically 30 km wide and 56 km long, its length parallel to the satellite ground track.
3) A water mask is added to the scene using only the 20 biggest objects from the SWBD [35] water mask database that are present in the scene. The hydrology target is always at the center of the scene.
4) For each object in the water mask we apply the following empirical algorithm (obtained by simplifying the statistical description of parameters of real data):
    a) The one at the center is always a river or a lake.
    b) It will be labeled as a lake if the length to width ratio is <40 or if its size is greater than 300 000 pixels (of 30 m side length). Otherwise, they are labeled as rivers.
    c) If it is a lake:
        i) Reflectance is mostly Lambertian, so a random value is chosen for the fraction of diffuse reflectance: $x_{\text{diff}} \in U(0.7, 1)$, where $U(a, b)$ is the uniform distribution in $[a, b]$.
        ii) The fraction of specularity is $(1 - x_{\text{diff}})$ and fwhm $\in 10^{U(\log 10(0.01), \log 10(0.1))}$.

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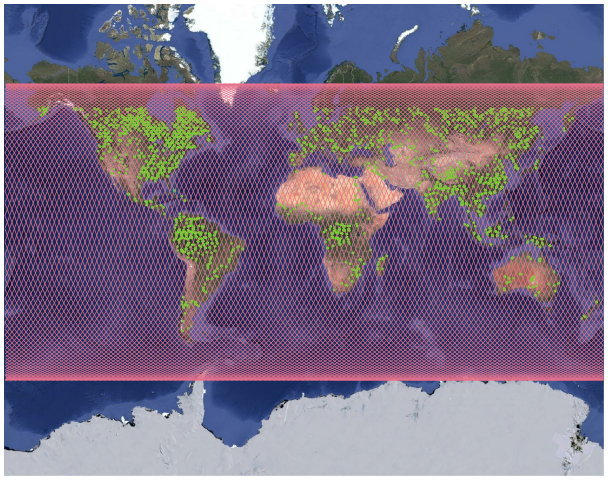


Fig. 7. Map showing as green dots the location of the virtual stations used. The grid shows the ground tracks of the Jason-3 mission.

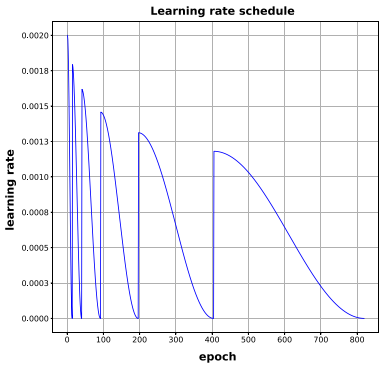


Fig. 8. Evolution of the learning rate using the cosine annealing with restarts and decay schedule with the following parameters: initial learning rate of 0.002, initial period of 13 epochs (NB: these are machine learning epochs, or computation cycles, completely unrelated to waveform epochs, τ), period multiplier of 2, restart decay of 0.9.

iii) The backscatter coefficient is $\sigma_0 \in U(15, 30)$ dB.

d) If it is a river:

i) fwhm $\in 10^{U(\log(0.01),\log(0.6))}$

ii) Reflectance is mostly specular, so $x_{\text{diff}} \in 10^{U(\log 10(0.00005),\log(0.02*\text{fwhm}))}$

iii) The backscatter coefficient is computed from the original measurement.

5) We generate several backscatter configurations for the same scene and cycle in the spirit of data augmentation [36], [37], although we simply regenerate the random coefficients as described above. We refer to the number of configurations as the augmentation factor.
6) The position of the altimeter will be given by its real trajectory, and we can then also use the real AGC and track range.
7) To add realism to the radargrams, a convolution of each simulated waveform with the instrument PTR (a sinc function) is made, creating saturation (clipped) peaks as observed in real data. However, to obtain a realistic proportion of saturated data, an iterative algorithm was implemented to obtain the correct saturation distribution.

To compare with *in situ* data, we used cycles 58 to 104 (september 2017 to december 2018). After compiling the database, we have 2517 virtual stations over the 47 cycles with an augmentation factor of 6, obtaining 709 794 different simulations. These were divided into a training dataset of ca. 570 000 simulations and a validation dataset of ca. 140 000 simulations.

B. Deep Convolutional NNs

As stated at the beginning of this section, our objective is to define a NN that takes as input a radargram, and gives as output the epoch corresponding to the water surface. This problem is hence one of finding the function $f : R^2 \to R_{\geq 0}$ that predicts one positive real value

$$f(\text{radargram}) = r = \begin{cases} 0, & \text{if no water is detected} \\ \text{water surface epoch}, & \text{otherwise.} \end{cases}$$

In machine learning terms, this problem is one of regression analysis. Because we can view the problem as one of finding a point in the radargram, our model is based on the ResNet-50 NN architecture, which performs well in image recognition [38]. Nevertheless, the model has to be adapted to our particular needs; ResNets are designed for classifying RGB color images, whereas in this study the input radargrams are like grayscale (or pseudocolor) images, and thus the input is just one matrix instead of three.

We trained a convolutional NN using PyTorch [39]. The 50-layer ResNet is an interesting compromise between the computation time and the time needed to finetune the hyperparameters in order to obtain satisfying results (hyperparameters are the parameters controlling the learning process as well as the topology and size of the NN). In particular, using 152 layers did not give better results than with 50 layers, and was significantly longer to train. On the other hand, an 18-layer network obtained almost as good results (92% accuracy versus 96% accuracy on the test dataset) while being faster to train. For this study we chose, however, to prioritize accuracy and use the 50-layer network.

We followed recommendations published in the literature to finetune the hyperparameters [40], [41]. The NN weights were initialized like in [42] because of the use of rectified linear units (ReLUs) in the network [43]. The optimizer was Adam [44]. The learning rate is one of the most crucial hyperparameters to finetune. It determines how much the NN weights are adjusted at each step. In order not to depend on a single value, a learning rate schedule that adjusts the learning rate during training was adopted. Empirically, we favored cyclical schedules [45], especially Loshchilov and Hutter's cosine annealing with restarts [46], [47], as seen in Fig. 8. We added a maximum value decay at every restart, because without it the loss function value rose when the learning rate restarted.

Also, our networks benefited from dropout [48] as long as the training database was too small (less than 500k examples): 25% of convolutional units were dropped, as well as 20% of the fully connected layer units. When the database became big enough, adding dropout slightly degraded the performance of the NN, whereas there was not much overfitting to prevent anymore. Hence, we chose to keep a slight overfitting

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TABLE III
PERFORMANCES SUMMARY

CPU (cores/threads)	GPU (memory)	maximum batch size	time/epoch (minutes)	time/sample (seconds)
Xeon E5-1650 3.6 GHz (6/12)	GTX 1080 (8 Go)	64	19.6	0.01
2x Xeon Silver 2.1 GHz 2x (12/12)	2x V100 (2x 32 Go)	512	5.5	0.02

to benefit from a better validation accuracy. Moreover, ResNet convolutional layers are paired with batch normalization layers. According to [49], batch normalization "acts as a regularizer, in some cases eliminating the need for dropout." We only kept 20% of dropout before the output fully connected layer. This did indeed enhance the performance.

Optimizing for the available hardware (GPU and memory), we trained our NNs with batch sizes varying from 64 to 512 samples. We did not notice significant differences in the prediction performance, although the use of bigger batches obviously reduced the training time. Table III summarizes the measured timings depending on the hardware configurations used during the study. Note that the processor used along with the NVIDIA GTX 1080 was faster than the one used with the V100. We could still save time when working with the V100 if we had a faster CPU.

V. EXPERIMENTAL RESULTS

As usual when working with deep learning, the simulations previously described were split into two groups: training and validation simulations. During the training phase, the labels were obtained from the simulation's scene. During the training phase, we strived for an accuracy of 90% within a range of ± 20 cm. That is, the result was deemed accurate enough if it was within the range label $\pm$ 20 cm. Real measurements were tested against the resulting NN to validate its real world performance.

A. Validation on Simulations

We created many different NNs to tackle our radargram analysis problem. All of these networks used the ResNet architecture, but differed in the NN weight initialization strategy, how the NN weights were updated (the optimizer), and the learning rate schedule. Fig. 9 shows the accuracy of the best model we found. This NN, used on the remaining of the article, achieved 90.75% of success. It was trained during 819 epochs on a 709 494-sample dataset (with 80% of the set used for training and 20% for validation). The learning rate schedule was the cosine annealing with restarts and decay shown on Fig. 8: 20% of the fully connected output layer units were dropped out.

B. Validation on In Situ Data

To test the performance of our model on real radargrams, we evaluated the results obtained against the values measured by *in situ* water level gauges. The regions chosen for validation depended mostly on the availability of such gauges, and either published by hydrologists or public agencies.

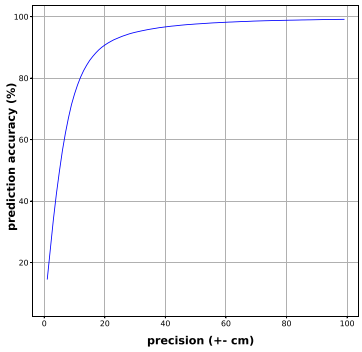


Fig. 9. Network performance on the validation set. This network reached 90.75% of correct predictions within a range of ± 20 cm on the validation set. If we wanted a precision better than 30 cm, network's prediction accuracy would be of 95%.

For each virtual station in the study domain, we processed the radargrams corresponding to successive Jason-3 flyovers with our best NN in order to build a water height time series. The comparison of this time series with the *in situ* water-level gauge time series permitted us to assess the performance of the NN. Most of the time, *in situ* time series are sampled every 15 min to 1 h. It can reach up to 12 h, depending on the region. We downsampled the *in situ* series to match the moments when the satellite flies over the hydrology target.

In situ gauges have to be located as close as possible to the studied virtual stations. However, it is difficult to find publicly available gauge data at the precise position of the virtual station. To avoid problems with different reference heights we compare anomaly time series instead of absolute water height. The height bias induced by the presence of the river slope is also partially removed when comparing water elevation anomalies. Anomaly time series were computed by removing the temporal means of the time series. Comparisons were made using the correlation coefficient and the RMSE following [50]. Note that the RMSE includes the uncertainties due to the satellite ground track jitter and noise from both the measure and the method.

1) Rio Negro (Brazil) (Fig. 10): Rio Negro is among the ten largest rivers in the world. An *in situ* gauge is available in Manaus harbor, under Jason-3 track 63 on the Rio Negro near its confluence with the Amazon, where the river width is about 2500 m. *In situ* data were provided by the Brazilian Water Agency ANA (http://www2.ana.gov.br).

The gauge is 10 km upstream from the virtual station. The river slope is known to hugely vary during the seasonal cycle and its width is much greater near the harbor than near the confluence. Moreover, the sampling rate of the *in situ* series is of 12 h only and there can be up to 6 h difference when comparing measurements. All these can explain the gap between the two series around June 2017. Nevertheless, the correlation between the *in situ* and the NN time series is 0.998 and the RMSE is 28.8 cm, having a comparable correlation but much better RMSE than when comparing *in situ* data and the OCOG time series (RMSE: 64.3 cm)

2) La Garonne (France) [Fig. 11(a)]: Garonne is a mid-latitude small size river, with rapid level variations. At the crossing with the Jason ground tracks, it is about 150 m wide. Observations collected at *in situ* gauges are

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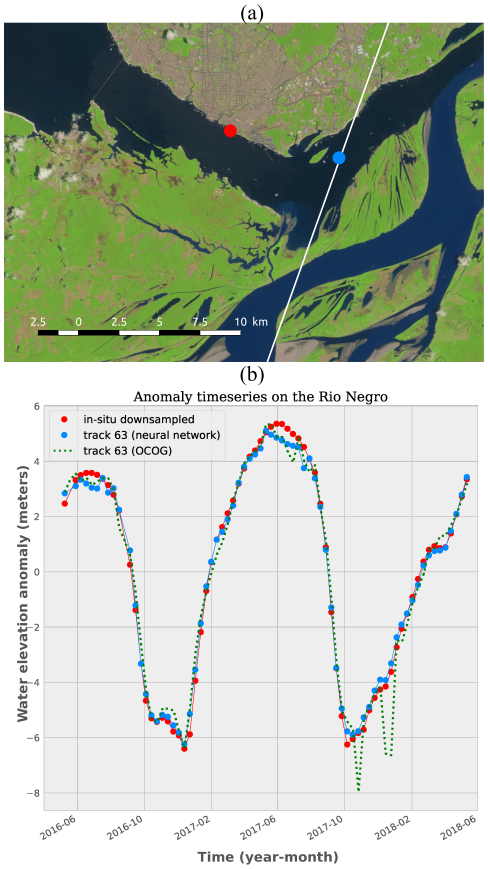


Fig. 10. (a) *In situ* gauge (red dot) and the virtual station (blue dot). The white line corresponds to Jason-3 ground track. (b) Jason-3 water elevation anomaly time series on the Rio Negro near Manaus (blue line and dots) compared to *in situ* anomaly time series (red line and dots) and the OCOG retracker (green dotted line). *In situ* series is downsampled to match the ten-day satellite sampling. Correlation NN/*in situ*: 0.998, RMSE: 28.8 cm. Correlation OCOG/*In situ*: 0.986, RMSE: 64.3 cm.

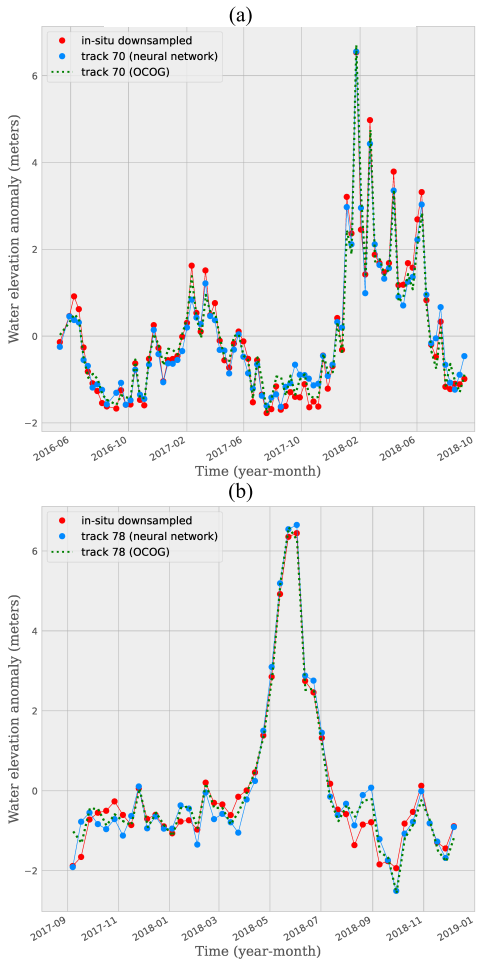


Fig. 11. Jason-3 water elevation anomaly time series (blue lines and dots) compared to *in situ* anomaly time series (red lines and dots) and the OCOG retracker time series (green dotted line). *In situ* series are downsampled to match the ten-day satellite sampling. (a) Garonne River near Marmande, France. Correlation NN/*in situ*: 0.985, RMSE: 30.3 cm. Correlation OCOG/*in situ*: 0.986, RMSE: 28.9 cm. (b) Pend Oreille River in northeastern Washington state, USA. Correlation NN/*in situ*: 0.983, RMSE: 36.6 cm. Correlation OCOG/*in situ*: 0.986, RMSE: 31 cm.

released publicly online by the SCHAPI (Service Central d'Hydrométéorologie et d'Appui à la Prévision des Inondations-http://www.hydro.eaufrance.fr). Water elevations are measured with a nonuniform time step, which depends on the water elevation variations. The median time step on the studied gauge is 30 min.

We compare the NN output for the target identified at the crossing with this river with an *in situ* gauge operating 6 km upstream (Marmande). The river is too narrow to be in the SWBD water mask. Consequently, the NN has never seen such rivers of this size during its training and it is an interesting challenge for our model. The correlation between *in situ* data and the NN time series is 0.985 and the RMSE is 30.3 cm. The OCOG time series provides comparable results with slightly beter RMSE (28.9 cm). We notice that the radar altimetry and the NN were able to follow the rapid rise in water elevation that occurred from February to June 2018.

3) Pend Oreille River (USA) [Fig. 11(b)]: We used the United States Geological Survey Water Data Bank (http://waterdata.usgs.gov) to find gauges close to Jason-3 virtual stations. One of them is on the Pend Oreille River, on a section whose width is 190 m, in a mountainous region of the state of Washington. The virtual station is 3 km downstream from the gauge. The *in situ* water elevation is measured every 15 min. The Pend Oreille region is hilly, the river width is not constant, and there is a lake a few kilometers away from the station. This contributes to disrupt the radar measure and creates more complex radargrams.

The correlation between the *in situ* data and the NN time series is 0.983 and the RMSE is 36.6 cm. The OCOG time series shows a correlation of 0.986 and RMSE of 31.

C. Validation on Satellite Ground-Track Crossovers

For a few ground-track crossovers, we can find two virtual stations on the same waterbody, on two different tracks. It means the satellite passes over the same river with two different points of view. Then we have two radargrams, which can be very different, although measured on the same object.

Among these stations, we were particularly interested in those that are overflown within a few hours only. Thus, we can compare the time series computed by the NN from the two virtual stations, presuming the water height has to be almost the same. In this article, we focused on two cases.

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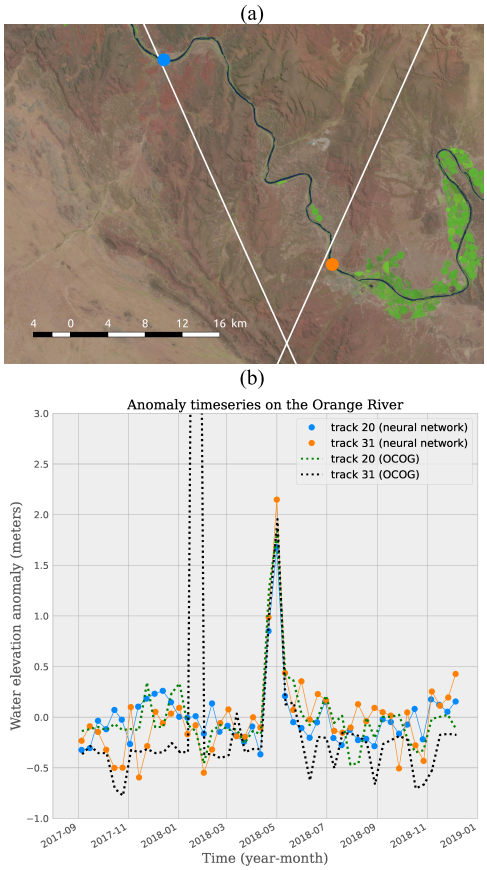


Fig. 12. (a) Orange River in South Africa. The satellite flies over the river twice within 10 h, on tracks 20 (blue dot) and 31 (orange dot). The two virtual stations are 37 km apart. The white line corresponds to Jason-3 ground track. (b) Jason-3 water elevation anomaly time series of the virtual stations on tracks 20 (blue line and dots) and 31 (orange line and dots). The OCOG time series are also shown for track 20 (green dotted line) and track 31 (black dotted line). Note that because of an outlier (of ordinate value ca. 12 m) for OCOG track 31 near 2018-02, the complete series is not shown. The correlation of the two NN series is 0.767 and the RMSE is 27.4 cm. The correlation of the two OCOG time series is 0.164 and the RMSE is 167.6 cm because of the presence of the outlier.

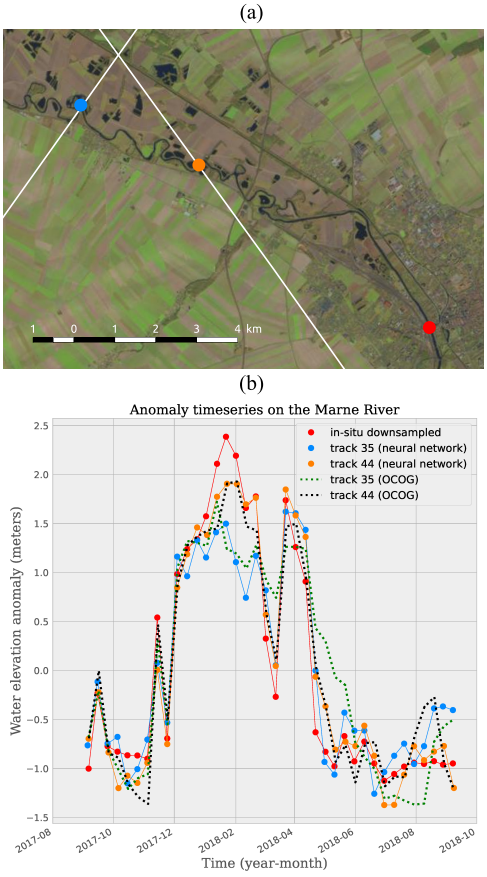


Fig. 13. (a) Marne River at Châlons-en-Champagne. The satellite flies over the river twice within 8 h, on tracks 35 (blue dot) and 44 (orange dot). The two virtual stations are 4 km apart and the *in situ* gauge is 9 km upstream the closest virtual station. The white line corresponds to Jason-3 ground track. (b) Jason-3 water elevation anomaly time series (blue and orange lines and dots) and *in situ* time series at Châlons-en-Champagne (red line and dots). The OCOG time series are also shown for track 35 (green dotted line) and track 44 (black dotted line). The two NN series have a correlation of 0.954 and an RMSE of 27.4 cm. The two OCOG time series have a correlation of 0.921 and an RMSE of 41.9 cm. The correlation between *in situ* data and track 35 is 0.938 for the NN and 0.899 for the OCOG time series. Their RMSE is 42.7 for the NN and 50.9 for OCOG. For track 44, the situation is very similar with correlation to *in situ* data of 0.975 for NN and 0.965 for OCOG and RMSE of 26 cm for NN and 30.9 cm for OCOG.

1) Orange River (South Africa) (Fig. 12): The region is a dry mountainous plateau and the river width varies between 50 and 200 m. There are 37 km between the two virtual stations, and are overflown at a 10-h interval. The correlation of the two NN series is 0.767 and the RMSE is 27.4 cm. The correlation of the corresponding OCOG time series is 0.164 and the RMSE is 167.6 cm. The correlation for OCOG is catastrophically low, and with a huge RMSE because of the presence of the outlier. But even if the outlier is ignored, the NN has much more consistent results for crossovers than the OCOG retracker.

2) La Marne (France) (Fig. 13): The case is particularly interesting since there is an *in situ* gauge near the crossover. Virtual station on track 44 (respectively, 35) is 9 km (respectively, 13.5 km) downstream from the gauge.

The river has many meanders and a mean width of only 50 m. Moreover, there are many small ponds close to the river, which is in an agricultural area. This can explain the differences between the three time series on Fig. 13. Because the three stations are not so close from each other, our model probably measures the dynamics between the river and its environment between the stations. Then, this contributes to degrade both the correlation coefficient and the RMSE.

The two NN series have a correlation of 0.954 and an RMSE of 27.4 cm. The two OCOG time series have a correlation of 0.921 and an RMSE of 41.9 cm. The correlation between *in situ* data and track 35 is 0.938 for the NN and 0.899 for the OCOG time series. Their RMSE is 42.7 for the NN and 50.9 for OCOG. For track 44, the situation is very similar with correlation to *in situ* data of 0.975 for NN and 0.965 for OCOG and RMSE of 26 cm for NN and 30.9 cm for OCOG. In this zone, the NN performs consistently better (15% or 5 cm) RMSE with respect to the OCOG time series.

A summary of the results is presented on Table IV (where "us" means upstream and "ds" means downstream).

D. Visualizing the Behavior of NN Components

The behavior of deep NNs is still subject to much scientific research work [51]. For a particular network, it is interesting to know how the network reacts to a specific input and,

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TABLE IV
SUMMARY OF VIRTUAL STATION VALIDATION FOR NN AND OCOG RETRACKERS

Reference or Crossover	River width	Distance from gauge (km)	Gauge sampling rate	Correlation NN	RMSE NN (cm)	Correlation OCOG	RMSE OCOG (cm)
In situ Rio Negro (BR)	4 km	10 (us)	12h	0.998	28.8	0.986	64.3
In situ Garonne (FR)	150 m	6 (us)	30'	0.985	30.3	0.986	28.9
In situ Pend Oreille (US)	190 m	3 (ds)	15'	0.983	36.6	0.986	31.0
In situ La Marne tr 35 (FR)	50 m	13.5 (ds)	30'	0.938	42.7	0.899	50.9
In situ La Marne tr 44 (FR)	50 m	9 (ds)	30'	0.975	26.0	0.965	30.9
Orange River tr 20/31 (SA)	50-200 m	37 km	10h	0.767	27.4	0.164	167.6
La Marne tr 44/35 (FR)	50 m	4.5 km	7h18'	0.954	36.7	0.921	41.9

us: upstream ; ds: downstream

more specifically, to find out what are the features in the radargram that the NN reacts to. We adapted Zeiler's [52] occlusion experiment to regression in order to visualize which radargram's features are important to our NN and which ones are taken into account to output the water surface elevation. For a given radargram having predicted an epoch τ_0 and converting this to water elevation W_e^0, the method consists in sliding a 3×3 occluding window centered on pixel p of the radargram and monitoring the NN's new output converted to water elevation, $W_e(p)$. The value on each pixel p of the heatmap is the difference $W_e(p) - W_e^0$, and is hence in meters.

For each position of the window we compute the difference between the NN prediction with occlusion and the original prediction, without occlusion. Once every position has been occluded, we have a heatmap of the essential radargram features.

Red pixels mean the NN overestimates the water elevation, whereas blue pixels mean it underestimates the elevation. If an area is white, it means that the NN does not give importance to this portion of the radargram. The stronger the color, the more important to the network is the associated radargram region.

From the heatmaps in Fig. 14 we can draw the following conclusions.

1) The trained convolutional NN is sensitive to the parabolas when they exist, or to the areas with a high power level. This is something we expected since this is where we visually find the information about the water surface elevation.
2) The NN gives a strong importance to the main waterbody, near the center of the radargram. Other waterbodies are ignored or given a low importance as seen on cases 1 and 2. The parabola under the centered one is the reflection of a lake we want to ignore. As seen on the heatmap, this lake is given an extremely low importance by the NN.
3) The network takes the saturation into account because it gives importance to the sidelobes (*sinc*, from the altimeter PTR), when they are visible. On case 1, there is a strong saturation. On the heatmap, the saturation peak is visible as well, meaning that the NN gives importance to it and uses the sidelobes to extract information about the water surface elevation.
4) Finally, on case 3, the satellite ground track is almost parallel to the river. Because of its meanders, there are many close encounters resulting in some waveforms having a higher signal power than the main central signal. The river slope is visible in the radargram: from waveforms 20 to 60, the river surface elevation increases. As expected, the NN focuses on the middle waveforms. When waveforms from 30 to 40 are hidden, the network overestimates the height. Indeed, these waveforms correspond to lower water levels. The exact opposite happens for waveforms from 40 to 50. For this radargram, the NN needs approximately 20 waveforms to estimate the elevation.

To conclude, our NN extracts the information from multiple waveforms, with a particular focus on the centered ones, and has learned how to deal with saturated waveforms. This is crucial in order to process Jason-3 hydrology targets worldwide, because more than 80% of hydrological targets produce saturated waveforms (for all the radargrams at our disposal, 80% of them had at least one saturated waveform).

VI. DISCUSSION

This article has shown an innovative method for retrieving hydrological time series from altimetry. We started by recalling the importance of constituting these time series, and the challenges to be tackled. The usage of the new DEM tracking mode onboard Jason-3 and Sentinel-3 satellites has radically increased the quantity of available data in hydrology contexts.

A brief presentation of the principles of radar altimetry allowed to make clear the difference between radargrams in ocean and in hydrology contexts; namely, the fitness of purpose of using the Brown model. In the literature, approaches to obtaining hydrology time series from radargrams use retracking algorithms based either on physical models of more complexity than the Brown type, or by heuristics ones that have proven useful, such as the OCOG/Ice-1 retracker. These classical approaches do not act on the radargram, but treat each waveform independently. In contrast, by restating the problem as an interpretation of the complete radargram, we can use deep learning techniques and extract information of the radargram as a whole.

A deep NN is defined by an architecture and the weights of its neurons. The architecture is chosen according to state-of-the-art knowledge (ResNet in our case), but many techniques are used to create the best weights when using a training dataset, using supervised learning. For doing so, we need

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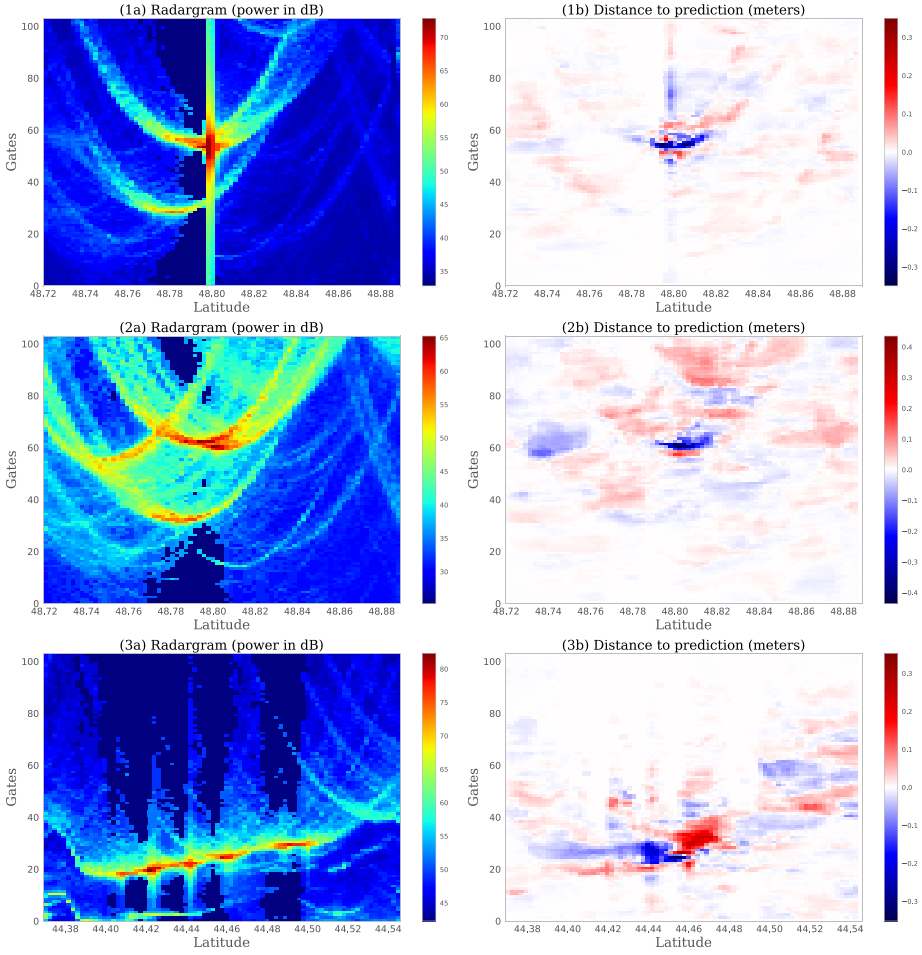


Fig. 14. Radargrams and their corresponding heatmaps. Using a sliding occlusion window, heatmaps show the effect of hiding a portion of the image on the predicted water elevation. See text for more details. (1a) and (1b) Pend Oreille River, USA. Cycle 66. (2a) and (2b) Pend Oreille River, USA. Cycle 102. (3a) and (3b) La Garonne, France. Cycle 90.

a set of radar altimetry waveforms and their "true" water height. With real data (115 782 virtual stations), we only have the radar waveforms, with no ground truth associated with these data. And even for those cases where we have the associated *in situ* data, the latter is not necessarily the truth, because these measurements are neither perfect, nor measure exactly the same object than altimetry does. For instance, the altimetry measurement and the *in situ* station can be several kilometers away, where the effect of the river slope increases the discrepancy. In this article we chose instead to use a very complete altimetry simulator to constitute a database of 710 000 realistic simulated radargrams, all based on the trajectories and ancillary data of real measurements. The scene used for the simulations merges several sources to establish a "ground truth" with which we can label the resulting radargrams. The scene uses topography from the SRTM-30 dataset [33] and water mask from SWBD [35]. The reflectance of the background land cover being much less than that of water, we describe it by a constant of very low reflectance. The objects of the water mask are more realistically produced by modeling both lake and river objects. The first ones are mostly diffuse, the latter mostly specular, but both are a mixture of these two. Random distributions are used to create the reflectance characteristics of these objects, while maintaining some constraints so as to make them physically plausible.

From a machine learning perspective, the problem we seek to solve is obtaining the water height from the radargram of a known target (whose position is near the center of the radargram). By iterating over all available radargrams over a given target, the NN constructs a water-level time series, point by point.

We used PyTorch [39] and obtained a fairly satisfactory NN for Jason-3 altimetry data, with 90.75% success on the validation dataset. When applied to real measurements, we compared five virtual stations to *in situ* gauge data, finding correlations of over 95% for all of them and RMSE between 26 and 43 cm. For two crossovers, where we checked internal consistency, we found correlations of 0.954 and 0.767 and RMSE of 36.7 and 27.4 cm for the NN time series of La Marne and the Orange River (respectively). The same crossovers with the OCOG time series fared very differently, the Orange River showing a very low 0.164 correlation coefficient, and an RMSE of 167 cm. The OCOG crossover correlation for the Marne was slightly worse than that of the NN (0.921) and with a higher RMSE (41.9 cm) (Table IV).

These correlations have very high values and correspond to very low time differences between the altimetry measurement

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and the gauge measurement (less than 30' except for the Rio Negro, which is the largest one and one that flows unhindered). In contrast, validation with track cross-overs over France and South Africa had time lags of 7 and 10 h and were in rivers subject to water management maneuvers (Gariep Dam in South Africa, Joinville over La Marne). In these cases, crossover correlations are not as good as those presented for *in situ* data, but remain nonetheless, worthwhile. In general, we can say that the performance of the NN was similar or better to that of the OCOG retracker, and that it did not fail in some particular cases, as shown when analyzing the crossover data of the Orange River. The NN seems to provide reliable and consistent time series that behave as well or better than the OCOG retracker.

The method was proven to be reliable for rivers as narrow as 50 m (La Marne); this could be surprising given the size of the footprint of the radar altimeter (several km). However, our method relies on the signature that is present on the radargram and not directly on river width. For very small rivers, the backscattered signal from the water will not be enough to manifest itself against the background. However, the return power can be sufficient to appear as a distinct signature in the radargram when either the specular component of the backscatter or the integrated energy of the return signal is big enough with respect to the background. Once the signature of the object is "imprinted" on the radargram the deep learning method deduces the water level based on the position of the parabola on the radargram.

We also tested the performance of the NN using partial occlusion of the radargrams, as a way of creating a heatmap of the features in the radargram that were more important for the regression output value. As expected, we found that the central features were given priority, and that waveform saturation was correctly taken into account, when present.

VII. Conclusion

With respect to the original target of creating an automatic processing method for retrieving water heights from altimetry data, we believe the progress shown in this article is substantial. A first takeaway message of this work is that there *is* information in the radargrams that can be extracted for hydrology. As with all machine learning techniques, creating a NN is cumbersome and computationally expensive but, once created, its use is very fast and efficient. Furthermore, the NN is able to use generalization, reacting appropriately to radargrams even when their kind has never been seen before. Using this NN makes possible the analysis of the hundreds of thousands of virtual stations available with the advent of satellite altimeters using the DEM tracking mode.

However, this is not a perfect solution at global scale; there is still work to be done. First, the final NN is very sensitive to the training dataset, which must exhibit a representative sampling of the data to be processed. The crux of our method is using our altimetry simulator (with which we can obtain appropriate labels for our training data) as a proxy of real altimetry data. It is nevertheless very difficult to create a perfectly realistic training database.

Second, there is a wealth of historical data from missions not using the DEM tracking mode. These missions show radargram variations that are more difficult to tackle with the present network: the tracker range can shift its position substantially when approaching the waterbodies (and it can even create a radargram made of echoes completely unrelated to them). As a first step using this methodology and restricting ourselves to the DEM tracking mode altimeters, this problem is simply ignored, but this problem should clearly be overcome to be able to exploit historical altimetry data.

Third, we noticed that the NN can infer a great number of hydrological time series where only a couple of points are outliers. In our approach we use the information of successive waveforms to create a radargram and obtain information from the successive waveforms, all of them from the same satellite flyover. However, we can go even further and use the information from successive radargrams, at different dates. Using this information, we believe these outliers can be detected and the general accuracy improved if the temporal information was included in the network (using, e.g., temporal convolutional networks [53]).

Finally, the results of the network are given with no explicit justification for them. In line with recent approaches for explainable artificial intelligence (AI) [54]–[56], we believe there are tools to be developed, the first ones being the heatmaps of Fig. 14 in Section V.

The NN obtained is very usable and completely automatic, giving good results when compared to what is obtained with manual processing.

Acknowledgment

The authors would like to thank Chedy Raïssi for a fruitful discussion at the beginning of this project and the French national altimetry data service Centre de Topographie des Océans et de l'Hydrosphère (CTOH) at the Laboratoire d'Études en Géophysique et Océanographie Spatiales (LEGOS) for providing access to the database of altimetry products, and the Brazilian Water Agency ANA for the *in situ* data of the Rio Negro.

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